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ALABAMA LICENSED No. 24744 PROFESSIONAL ENGINEER ROGER JAMES SMITH
2/20/2024

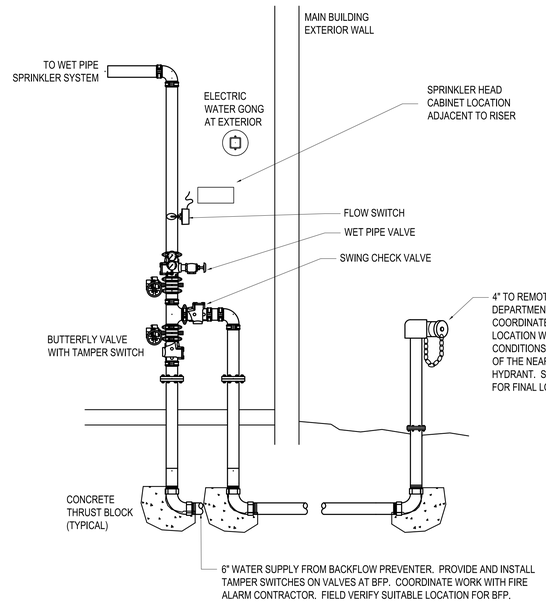
ISSUE	DATE	DESCRIPTION

GENERAL SHEET NOTES

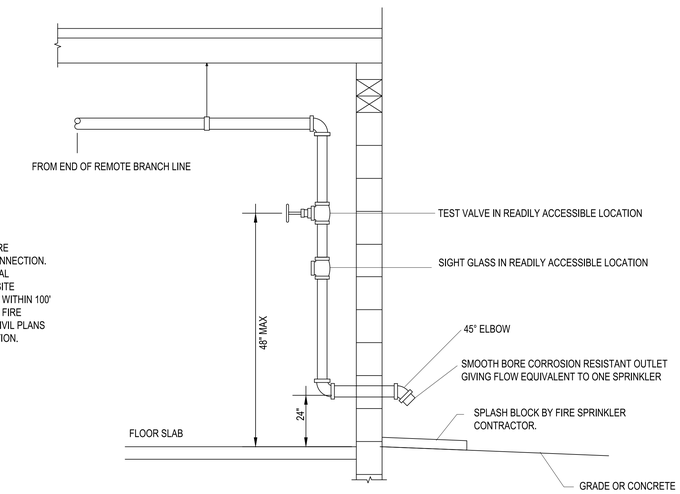
1. TAMPER AND FLOW SWITCHES BY SPRINKLER CONTRACTOR. FIRE ALARM CONTRACTOR SHALL PROVIDE AND INSTALL ALL NECESSARY APPURTENANCES, DEVICES, ALARMS, TO AUTOMATICALLY CALL THE FIRE DEPARTMENT UPON ALARM. THEY SYSTEM PROVIDED AND INSTALLED SHALL BE A FULLY FUNCTIONING AUTOMATIC SYSTEM. THE FIRE SPRINKLER CONTRACTOR IS RESPONSIBLE FOR PROVIDING ANY RELATED POWER WIRING OR CONDUIT REQUIRED TO MAKE THE FIRE SPRINKLER/FIRE ALARM SYSTEM FUNCTIONAL. ALL ELECTRICAL WORK SHALL BE INSTALLED PER THE LATEST NEC REQUIREMENTS. FIRE ALARM GONG SHALL BE ELECTRIC.
2. PROVIDE WET PIPE SPRINKLER PROTECTION PER NFPA 13 FOR THE ENTIRE BUILDING, BELOW THE CEILING FOR THE INTERIOR CONDITIONED SPACE ON THE FIRST AND SECOND FLOOR AS WELL AS THE HEATED APPARATUS BAY. PROVIDE SIDEWALL COVERAGE FOR ATTACHED COVERED PORCHES.
3. COORDINATE EXACT LOCATION OF ALL SPRINKLERS WITH THE CEILING AND LIGHTING LAYOUT.
4. LIGHT FIXTURES AND HVAC DIFFUSERS TAKE PRECEDENCE. ADD ADDITIONAL SPRINKLERS AS REQUIRED TO MEET "COVERAGE REQUIREMENTS".
5. IN MECHANICAL AND JANITOR'S ROOMS FINAL LOCATION OF SPRINKLERS SHALL BE DETERMINED AFTER EQUIPMENT AND DUCTWORK ARE IN PLACE. CONTRACTOR SHALL PROVIDE ADDITIONAL SPRINKLERS, IF NECESSARY, TO PROVIDE ADEQUATE COVERAGE IN ACCORDANCE WITH NFPA 13.
6. PROVIDE A LISTED GUARD FOR SPRINKLERS IN LOCATIONS SUBJECT TO MECHANICAL INJURY. THESE AREAS SHALL INCLUDE MECHANICAL ROOMS, ELECTRICAL ROOMS, UNDER STAIRWELL LANDING.
7. ROUTE SPRINKLER PIPING WITHIN THE TRUSS SPACE.TO THE GREATEST EXTEND POSSIBLE.
8. REFERENCE ARCHITECTURAL PLANS FOR CEILING TYPES AND HEIGHTS. PROVIDE COVERAGE PER NFPA 13 ACCORDINGLY.
9. SUBMIT ENGINEER STAMPED DRAWINGS FOR APPROVAL BY THE LOCAL GOVERNING AUTHORITY PRIOR TO BEGINNING ANY WORK.
10. PROVIDE CONCEALED HEADS IN LOCATIONS WITH LAY-IN OR GYPSUM CEILINGS. ESCUTCHEONS OR COVERS SHALL MATCH THE CEILING COLOR.
11. ALL WORK SHALL BE INSTALLED IN COMPLIANCE WITH THE REQUIREMENTS OF NFPA 13. ALL PIPING EXPOSED TO VIEW SHALL BE PAINTED TO MATCH SURROUNDINGS.

CALCULATIONS REQUIREMENTS

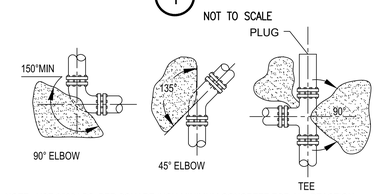
1. THE SYSTEM SHALL BE HYDRAULICALLY DESIGNED WITH A HOSE STREAM ALLOWANCE OF 250 GPM FOR LIGHT HAZARD AND 500 GPM FOR ORDINARY HAZARD AND DENSITY VALUES AS FOLLOWS:
   - LIGHT HAZARD DENSITY = 0.10 GPM/SF OVER THE MOST DEMANDING 1500 SQ. FT. WITH 225 SQ. FT. MAX COVERAGE FOR SPRINKLERS.
   - ORDINARY HAZARD GROUP 1 DENSITY = 0.15 GPM/SF OVER THE MOST DEMANDING 1500 SQ. FT. WITH 130 SF MAX COVERAGE FOR SPRINKLERS.
2. PROVIDE SHOP DRAWING AND CALCULATIONS:
   - ALL PIPING LABELED WITH REFERENCE TO HYDRAULIC CALCULATIONS.
   - PROVIDE QUALITY, MANUFACTURE, MODEL#, RATING, ORIFICE SIZE OF ALL SPRINKLER HEADS PROVIDED LIST ON SHOP DRAWING.
   - PIPE TYPE.
   - REMOTE AREA LOCATION.
   - HANGER DETAILS
   - HAZARD CLASSIFICATION
   - FLOW DATA.
3. SEISMIC NOTE: THE AREA SEISMIC REQUIREMENTS ARE MEET USING STANDARD NFPA 13 SUPPORTS.



1 WET PIPE FIRE RISER DETAIL
NOT TO SCALE

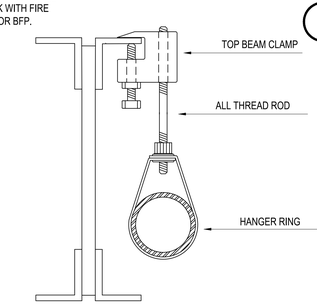


2 INSPECTOR'S TEST CONNECTION DETAIL - WET PIPE
NOT TO SCALE

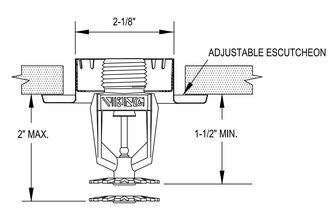


NOTE: USE 2500 LBS. PER SQUARE INCH MINIMUM CONCRETE FOR ALL REACTION BLOCKS. PROVIDE VISQUEEN OR FELT BETWEEN THE CONCRETE AND THE PIPE.

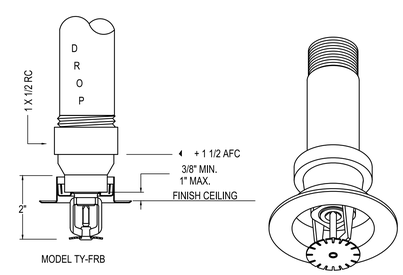
3 TYPICAL THRUST BLOCK DETAIL
NOT TO SCALE



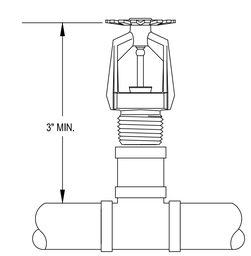
4 TOP BEAM CLAMP DETAIL
NOT TO SCALE



5 SPRINKLER HEAD DETAIL
NOT TO SCALE



6 RECESSED SPRINKLER HEAD DETAIL
NOT TO SCALE



7 UPRIGHT SPRINKLER HEAD DETAIL
NOT TO SCALE

A NEW CENTRAL STATION FOR
CITY OF SPANISH FORT FIRE RESCUE
7361 SPANISH FORT BLVD
SPANISH FORT, ALABAMA 36527

PROJECT NO: 23-001

SHEET NO:
FIRE SPRINKLER DETAILS AND NOTES

F100

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